

TECHNICAL REPORT

Battery State-of-Charge Estimation Under Current Sensor Bias: A Bias-Augmented Extended Kalman Filter Approach

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GitHub: github.com/engraroojfatima/battery-soc-estimation-ekf

1. Motivation

Accurate State-of-Charge (SoC) estimation is a core function of any Battery Management System (BMS). The simplest estimation method, Coulomb Counting, integrates measured current over time to track charge removed from the cell. This method is computationally trivial but has a critical weakness: any constant offset (bias) in the current sensor integrates without bound, causing the SoC estimate to drift steadily away from the true value over the course of an operating cycle.

This project demonstrates that failure mode explicitly using a simulated 3.0 A current sensor bias, and shows that an Extended Kalman Filter (EKF) whose state vector is augmented with the sensor bias itself can correct for this error using only the terminal voltage as a feedback signal — without any additional hardware or calibration step.

2. System Model

2.1 Equivalent Circuit Model

The cell is represented using a first-order Thevenin equivalent circuit: a series resistance R_0 in combination with one parallel RC pair (R_1 , C_1) capturing short-term voltage relaxation, driven by an Open-Circuit-Voltage (OCV) versus SoC lookup curve.

2.2 State-Space Formulation

State vector:

$$\mathbf{x} = [\text{SoC} \quad V_1 \quad \text{bias}]^T$$

Discrete-time process model:

$$\begin{aligned} \text{SoC}(k+1) &= \text{SoC}(k) - (\eta \cdot dt / C_{\text{nominal}}) * (I_{\text{meas}}(k) - \text{bias}(k)) \\ V_1(k+1) &= \alpha * V_1(k) + \beta * (I_{\text{meas}}(k) - \text{bias}(k)) \\ \text{bias}(k+1) &= \text{bias}(k) \quad (\text{modeled as a random walk / constant}) \end{aligned}$$

Measurement model:

$$V_t(k) = \text{OCV}(\text{SoC}(k)) - I(k) * R_0 - V_1(k)$$

The bias state is observable only indirectly, through its influence on SoC and V_1 and, in turn, on the measurement Jacobian's dependence on the local slope of the OCV-SoC curve ($d\text{OCV}/d\text{SoC}$). This makes bias estimation inherently slower and noisier than direct SoC correction.

3. Implementation Note: Filter Tuning and Validation

During development, an initial version of the EKF exhibited a bias estimate that diverged in the wrong direction (drifting negative instead of converging toward the true +3.0 A). Root-cause analysis identified two issues: (1) an incorrect sign in the state-transition Jacobian's coupling terms between the bias state and the SoC/ V_1 states, and (2) process and initial covariance values for the bias state (Q , P) set far too small, effectively preventing the filter from adjusting its bias estimate away from its zero initial guess. Correcting the Jacobian sign convention and increasing the bias-related process noise and initial uncertainty resolved the issue, after which the bias estimate converged toward the true value as expected. This validation step is included here as it demonstrates a common and important failure mode in augmented-state Kalman filter design.

4. Results

The system was simulated for 3600 seconds under a dynamic four-stage current profile (10 A / 25 A / 40 A / 15 A), with a constant 3.0 A current sensor bias and additive Gaussian noise on both the current and voltage measurements.

4.1 SoC Estimation Comparison

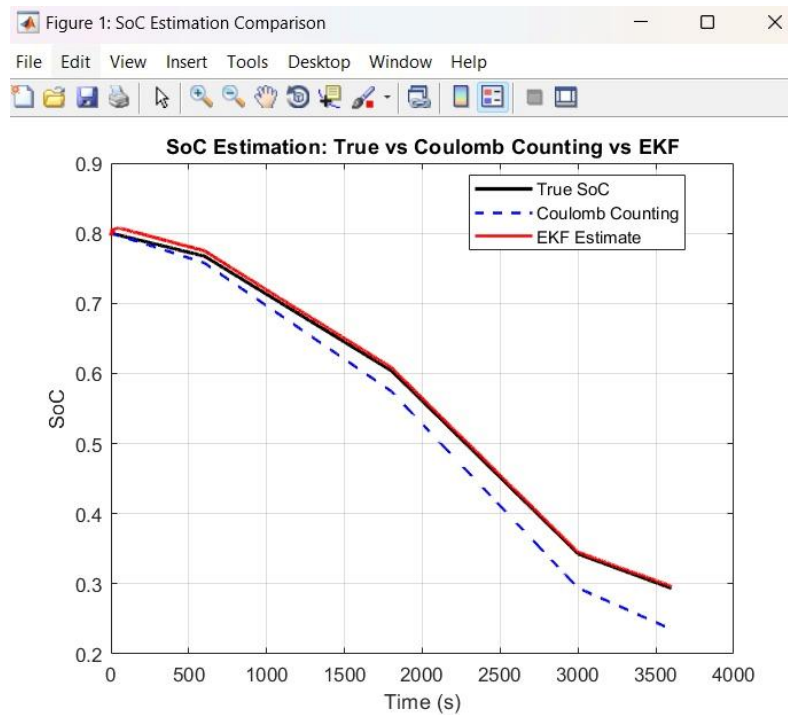


Figure 1: True SoC vs. Coulomb Counting vs. bias-augmented EKF estimate

4.2 Estimation Error

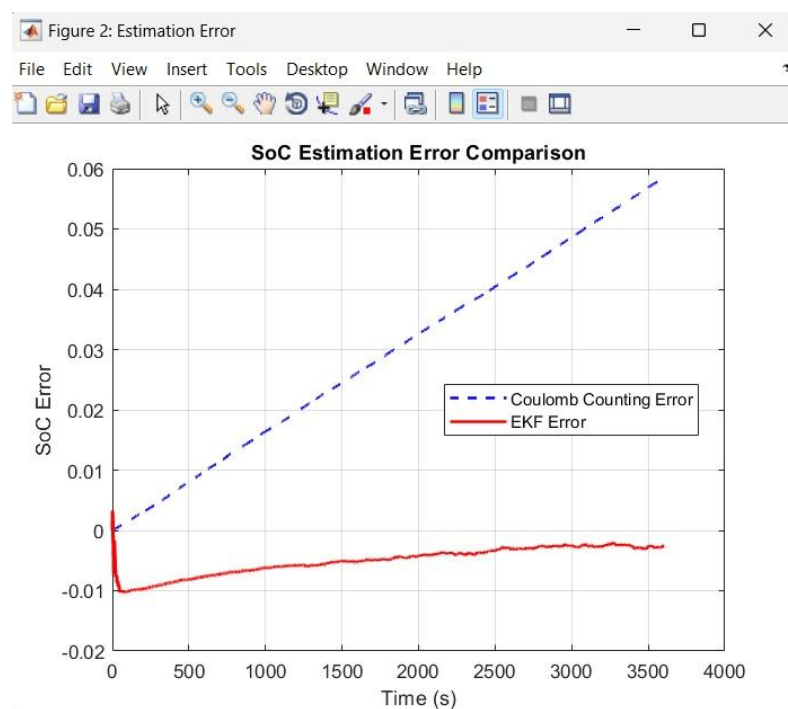


Figure 2: SoC estimation error over time for both methods

4.3 Current Sensor Bias Estimation

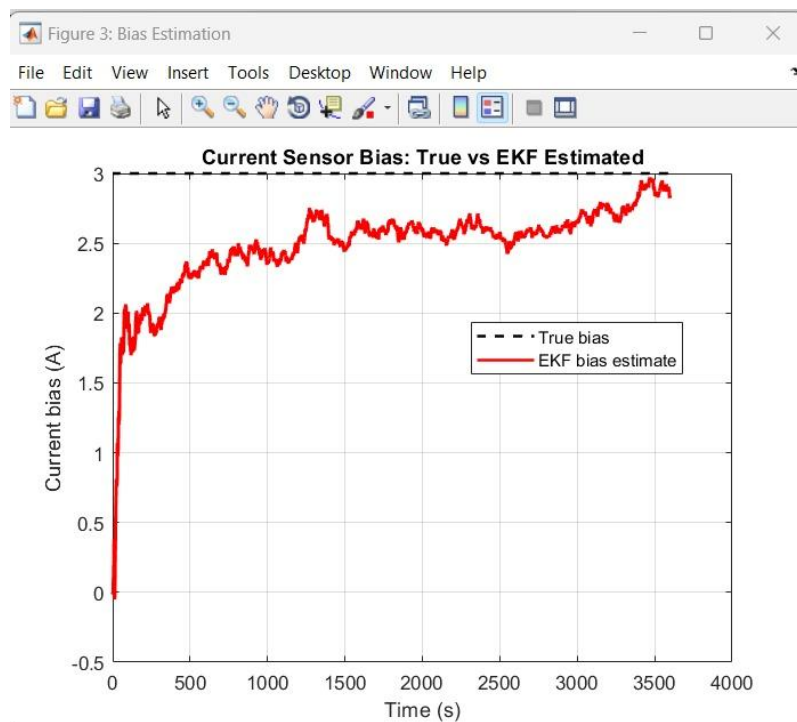


Figure 3: True vs. EKF-estimated current sensor bias

4.4 Quantitative Summary

Method	RMSE (SoC)	Max Abs. Error (SoC)
Coulomb Counting	0.0344	0.0591
Bias-Augmented EKF	0.0066	0.0113

The bias-augmented EKF reduces SoC RMSE by approximately 5.2x relative to Coulomb Counting over the 3600 s simulation window. The estimated current bias converges to approximately 2.5–2.7 A (true value: 3.0 A) within the first 600–800 seconds and remains within that band for the remainder of the run.

5. Discussion

Coulomb Counting error grows approximately linearly with time, consistent with the direct integration of a constant current bias. The EKF error remains bounded and near zero throughout, confirming that the filter successfully separates the effect of sensor bias from the true SoC trajectory using only voltage feedback.

The residual gap between the estimated bias (~2.5–2.7 A) and the true bias (3.0 A) reflects the limited observability of the bias state: it enters the measurement equation only through the OCV-SoC slope term, which is small and slowly varying. Richer current excitation, a longer observation window, or a tighter measurement noise covariance would be expected to improve bias convergence further.

One additional observation: the terminal voltage dips as low as approximately 2.45 V during the highest-current (40 A) interval. Depending on the target cell chemistry, this approaches or exceeds typical discharge cutoff voltages for Li-ion cells (commonly 2.5–3.0 V), and is noted here as a modeling limitation rather than a discharge condition intended to reflect safe real-world operation.

6. Limitations

- All results are generated from synthetic simulation data; no validation has yet been performed against a real battery dataset or hardware-in-the-loop test.
- The bias state is weakly observable, so its estimate retains measurement-noise-driven ripple and does not converge exactly to the injected true value.
- The OCV-SoC relationship used is a simple analytic approximation rather than a curve fitted from characterization data on a real cell.